

Semantic Exploration Comparison: Human-AI vs Human-Human vs AI-AI

Cross-Condition Analysis of Lag-Based Semantic Distance

PENPAL Project

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1 Overview

The semantic exploration outputs now use the **current lag-based structure**:

- **k**: lag
- **distance**: mean semantic distance for that lag
- **agent**: "interleaved" for the full exchange sequence, or **author_1** / **author_2** for speaker-specific trajectories

The processed filename is still **semantic_exploration_binned.parquet**, but the current generator computes **lag-based distances**, not the older non-overlapping window-centroid bins.

This notebook therefore no longer assumes old turn-level exploration tables. Instead it analyses:

1. **Interleaved sequence exploration** across conditions
2. **Speaker-specific exploration** using recovered role metadata

Important comparison note:

- **Slot 1** / **Slot 2** is the directly comparable cross-condition axis.
- **Human** / **AI** in Human-AI and **Starter-side** / **Non-starter-side** in same-type conditions are only used as within-condition descriptive views.

Primary cross-condition object. Per story, we compute the trapezoidal area under each slot’s mean distance(k) curve over the intersection of k values present for both slots, and take the unsigned difference:

$$A_{\text{semexp}}(i) = | \text{AUC}_{\text{slot}=1, i} - \text{AUC}_{\text{slot}=2, i} |.$$

This is label-invariant and comparable across conditions. The lag-binned LMM slot contrast and the log-distance slope view are retained as supplemental. Starter-mechanism and Human-AI speaker-type decompositions are deferred.

2 Load Data

Table 1: Loaded semantic exploration rows by condition and agent view

condition	agent	n_stories	n_rows	min_k	max_k
Human-AI	author_1	48	383	1	8
Human-AI	author_2	48	383	1	8
Human-AI	interleaved	48	480	1	10
Human-Human	author_1	31	248	1	8
Human-Human	author_2	31	248	1	8
Human-Human	interleaved	31	310	1	10
AI-AI	author_1	39	312	1	8
AI-AI	author_2	39	312	1	8
AI-AI	interleaved	39	390	1	10

3 Preprocessing

Interleaved rows: 1180

Speaker rows: 1886

4 Primary: Per-Story AUC Asymmetry

Table 2: Stories with per-story AUC computable in both slots

condition	n_stories
Human-AI	48
Human-Human	31
AI-AI	39

Table 3: Per-story $|\text{AUC}(\text{slot } 1) - \text{AUC}(\text{slot } 2)|$ by condition

condition	n	mean_A	sd_A	median_A
Human-AI	48	0.248	0.187	0.188
Human-Human	31	0.213	0.175	0.186
AI-AI	39	0.215	0.161	0.181

One-way ANOVA: A_semexp ~ condition

Df Sum Sq Mean Sq F value Pr(>F)

```

## condition      2  0.033 0.01655  0.536  0.586
## Residuals    115  3.549 0.03086
##
## Kruskal-Wallis (robust): A_semexp ~ condition
##
## Kruskal-Wallis rank sum test
##
## data: A_semexp by condition
## Kruskal-Wallis chi-squared = 1.0149, df = 2, p-value = 0.602
##
## Pairwise Wilcoxon (BH-adjusted):
##
## Pairwise comparisons using Wilcoxon rank sum exact test
##
## data: df_auc_stories$A_semexp and df_auc_stories$condition
##
##           Human-AI Human-Human
## Human-Human 0.77      -
## AI-AI       0.77      0.78
##
## P value adjustment method: BH

```

Per-story semantic-exploration AUC asymmetry by condition

$$A_semexp(i) = |AUC_{\{slot\ 1\}}(i) - AUC_{\{slot\ 2\}}(i)|$$

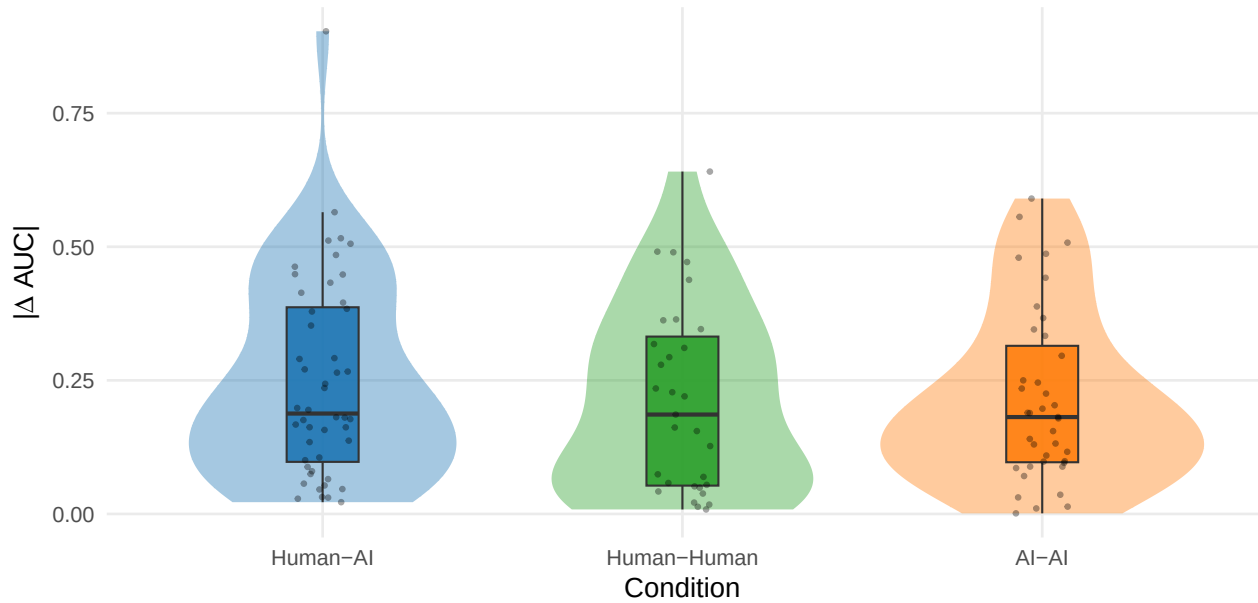


Table 4: Secondary: per-story $|\log(\text{distance}) \sim \log(k) \text{ slope diff}|$ by condition

condition	n	mean_A	median_A
Human-AI	48	0.087	0.072
Human-Human	31	0.098	0.083
AI-AI	39	0.078	0.067

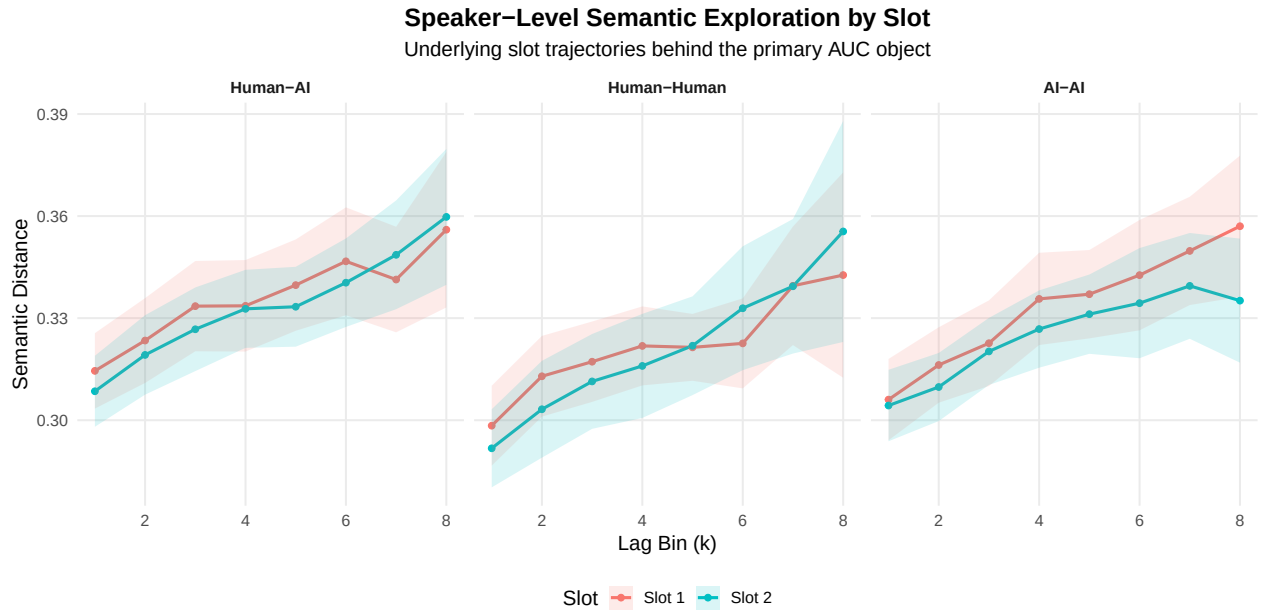
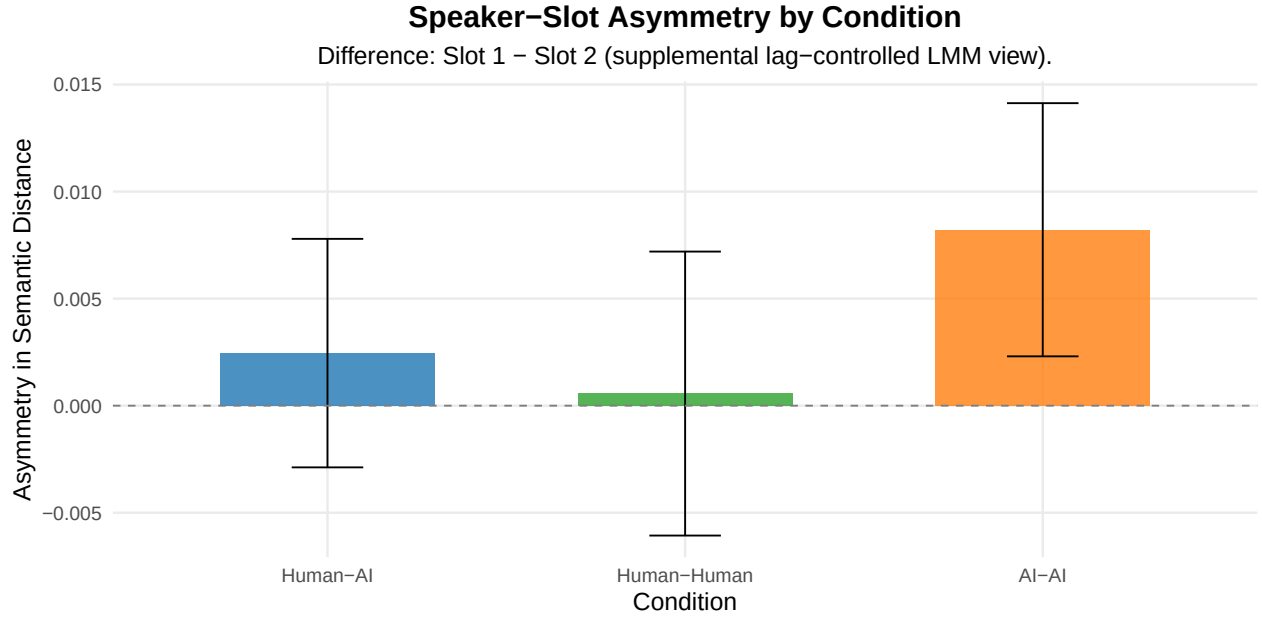
5 Supplemental: Lag-Binned Slot Contrast

```
## Speaker-slot asymmetry model:

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: distance ~ k + slot * condition + (1 | conversation_id)
## Data: df_speaker
##
## REML criterion at convergence: -6667.9
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -4.3647 -0.5761 -0.0185  0.5536  5.8438
##
## Random effects:
## Groups             Name             Variance Std.Dev.
## conversation_id (Intercept) 0.0009049 0.03008
## Residual                  0.0014194 0.03768
## Number of obs: 1886, groups: conversation_id, 118
##
## Fixed effects:
##
##              Estimate Std. Error      df t value
## (Intercept)    3.086e-01  5.045e-03 1.732e+02  61.161
## k              6.097e-03  3.789e-04 1.764e+03  16.090
## slotSlot 2    -2.455e-03  2.723e-03 1.764e+03  -0.902
## conditionHuman-Human -1.394e-02  7.581e-03 1.364e+02 -1.839
## conditionAI-AI    -2.628e-03  7.093e-03 1.364e+02 -0.370
## slotSlot 2:conditionHuman-Human 1.889e-03  4.343e-03 1.764e+03  0.435
## slotSlot 2:conditionAI-AI    -5.761e-03  4.063e-03 1.764e+03 -1.418
##
##              Pr(>|t|)
## (Intercept)    <2e-16 ***
## k              <2e-16 ***
## slotSlot 2      0.3673
## conditionHuman-Human 0.0681 .
## conditionAI-AI    0.7116
## slotSlot 2:conditionHuman-Human 0.6636
## slotSlot 2:conditionAI-AI    0.1564
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) k      sltSl2 cndH-H cAI-AI sS2:H-
## k              -0.337
## slotSlot 2    -0.270  0.000
## cndtnHmn-Hm  -0.590 -0.001  0.180
## condtnAI-AI  -0.630 -0.001  0.192  0.420
## sltS2:cnH-H   0.169  0.000 -0.627 -0.286 -0.120
## sltS2:AI-AI   0.181  0.000 -0.670 -0.120 -0.286  0.420
```

Table 5: Speaker-slot asymmetry in semantic exploration by condition

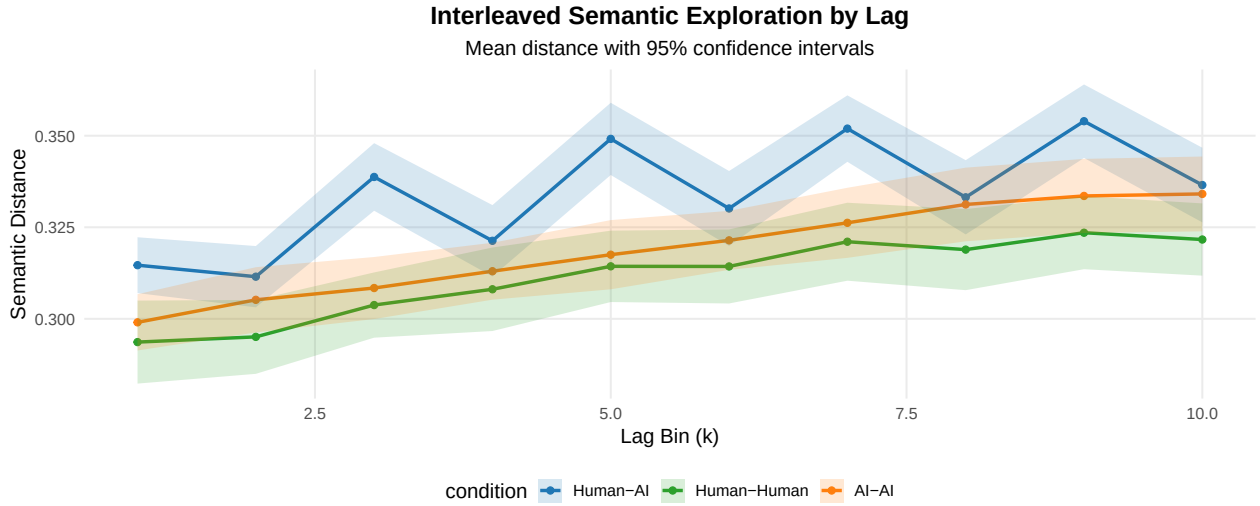
condition	difference	se	t	p
Human-AI	0.002	0.003	0.902	0.367
Human-Human	0.001	0.003	0.167	0.867
AI-AI	0.008	0.003	2.724	0.007



6 Supplemental: General Condition-Level Comparison

Table 6: Interleaved semantic distance by lag and condition

condition	k	mean_distance	sd_distance	n
Human-AI	1	0.315	0.027	48
Human-AI	2	0.312	0.030	48
Human-AI	3	0.339	0.033	48
Human-AI	4	0.321	0.034	48
Human-AI	5	0.349	0.035	48
Human-AI	6	0.330	0.036	48
Human-AI	7	0.352	0.032	48
Human-AI	8	0.333	0.036	48
Human-AI	9	0.354	0.035	48
Human-AI	10	0.337	0.036	48
Human-Human	1	0.294	0.032	31
Human-Human	2	0.295	0.029	31
Human-Human	3	0.304	0.025	31
Human-Human	4	0.308	0.032	31
Human-Human	5	0.314	0.028	31
Human-Human	6	0.314	0.029	31
Human-Human	7	0.321	0.030	31
Human-Human	8	0.319	0.031	31
Human-Human	9	0.324	0.028	31
Human-Human	10	0.322	0.028	31
AI-AI	1	0.299	0.024	39
AI-AI	2	0.305	0.028	39
AI-AI	3	0.308	0.027	39
AI-AI	4	0.313	0.025	39
AI-AI	5	0.318	0.030	39
AI-AI	6	0.321	0.026	39
AI-AI	7	0.326	0.030	39
AI-AI	8	0.331	0.032	39
AI-AI	9	0.334	0.032	39
AI-AI	10	0.334	0.033	39



Interleaved model summary:

Linear mixed model fit by REML. t-tests use Satterthwaite's method [

```

## lmerModLmerTest]
## Formula: distance ~ k * condition + (1 | conversation_id)
## Data: df_interleaved
##
## REML criterion at convergence: -5905.5
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.75893 -0.60448  0.01588  0.61147  2.89332
##
## Random effects:
## Groups           Name          Variance Std.Dev.
## conversation_id (Intercept) 0.0007415 0.02723
## Residual                0.0002656 0.01630
## Number of obs: 1180, groups: conversation_id, 118
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    3.161e-01  4.246e-03  1.458e+02  74.451 < 2e-16 ***
## k              3.268e-03  2.590e-04  1.059e+03  12.619 < 2e-16 ***
## conditionHuman-Human -2.356e-02  6.778e-03  1.458e+02  -3.476 0.000671 ***
## conditionAI-AI      -1.956e-02  6.342e-03  1.458e+02  -3.084 0.002441 **
## k:conditionHuman-Human 1.607e-04  4.134e-04  1.059e+03   0.389 0.697554
## k:conditionAI-AI      8.034e-04  3.868e-04  1.059e+03   2.077 0.038036 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) k      cndH-H  cAI-AI  k:cH-H
## k              -0.335
## cndtnHmn-Hm    -0.626  0.210
## condtnAI-AI    -0.670  0.225  0.419
## k:cndtnHm-H    0.210 -0.626 -0.335 -0.141
## k:cndtAI-AI    0.225 -0.670 -0.141 -0.335  0.419
##
## Condition contrasts (interleaved view):
## contrast              estimate      SE df t.ratio p.value
## (Human-AI) - (Human-Human)  0.02268 0.00639 115   3.552  0.0017
## (Human-AI) - (AI-AI)        0.01514 0.00597 115   2.534  0.0378
## (Human-Human) - (AI-AI)     -0.00754 0.00667 115  -1.130  0.7822
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: bonferroni method for 3 tests

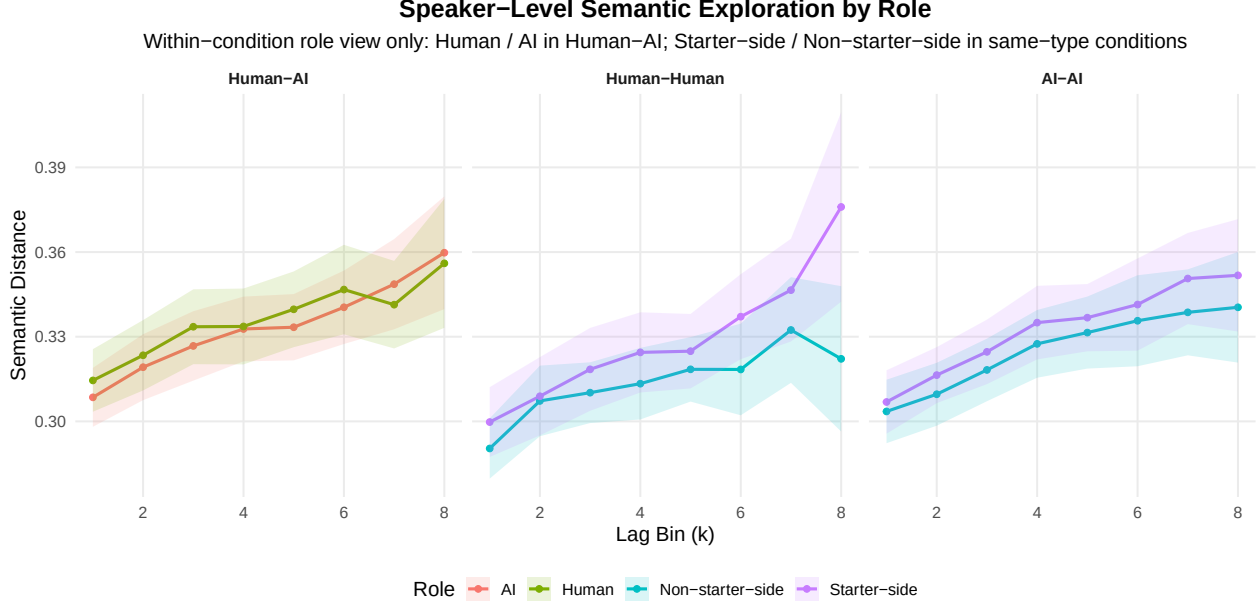
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7 Supplemental: Role-Level View

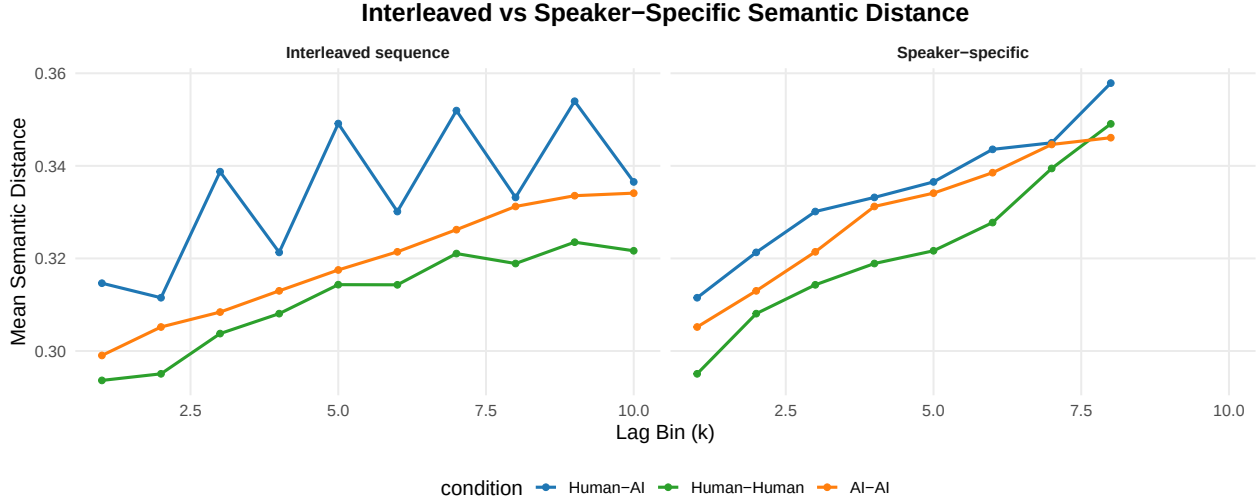
Table 7: Speaker-level semantic distance by condition, role, and lag

condition	condition_role	k	mean_distance	n
Human-AI	AI	1	0.309	48

condition	condition_role	k	mean_distance	n
Human-AI	AI	2	0.319	48
Human-AI	AI	3	0.327	48
Human-AI	AI	4	0.333	48
Human-AI	AI	5	0.333	48
Human-AI	AI	6	0.340	48
Human-AI	AI	7	0.349	48
Human-AI	AI	8	0.360	47
Human-AI	Human	1	0.315	48
Human-AI	Human	2	0.323	48
Human-AI	Human	3	0.334	48
Human-AI	Human	4	0.334	48
Human-AI	Human	5	0.340	48
Human-AI	Human	6	0.347	48
Human-AI	Human	7	0.341	48
Human-AI	Human	8	0.356	47
Human-Human	Non-starter-side	1	0.290	31
Human-Human	Non-starter-side	2	0.307	31
Human-Human	Non-starter-side	3	0.310	31
Human-Human	Non-starter-side	4	0.313	31
Human-Human	Non-starter-side	5	0.318	31
Human-Human	Non-starter-side	6	0.318	31
Human-Human	Non-starter-side	7	0.332	31
Human-Human	Non-starter-side	8	0.322	31
Human-Human	Starter-side	1	0.300	31
Human-Human	Starter-side	2	0.309	31
Human-Human	Starter-side	3	0.318	31
Human-Human	Starter-side	4	0.324	31
Human-Human	Starter-side	5	0.325	31
Human-Human	Starter-side	6	0.337	31
Human-Human	Starter-side	7	0.347	31
Human-Human	Starter-side	8	0.376	31
AI-AI	Non-starter-side	1	0.304	39
AI-AI	Non-starter-side	2	0.310	39
AI-AI	Non-starter-side	3	0.318	39
AI-AI	Non-starter-side	4	0.327	39
AI-AI	Non-starter-side	5	0.331	39
AI-AI	Non-starter-side	6	0.336	39
AI-AI	Non-starter-side	7	0.339	39
AI-AI	Non-starter-side	8	0.340	39
AI-AI	Starter-side	1	0.307	39
AI-AI	Starter-side	2	0.316	39
AI-AI	Starter-side	3	0.325	39
AI-AI	Starter-side	4	0.335	39
AI-AI	Starter-side	5	0.337	39
AI-AI	Starter-side	6	0.341	39
AI-AI	Starter-side	7	0.351	39
AI-AI	Starter-side	8	0.352	39



8 Supplemental: Interleaved vs Speaker View



9 Summary

Primary cross-condition result: per-story $A_{\text{semexp}} = |AUC_{\text{slot 1}} - AUC_{\text{slot 2}}|$ from trapezoidal integration of the distance(k) curve over the intersected k range. Tested with ANOVA + Kruskal and pairwise BH-adjusted Wilcoxon. A secondary log-distance-slope asymmetry is reported for dynamic-shape continuity with the paper.

Supplemental views retained:

- lag-controlled slot-contrast LMM ($\text{distance} \sim k + \text{slot} * \text{condition}$)
- interleaved sequence model and condition descriptives
- within-condition role curves (Human / AI in Human-AI; Starter-side / Non-starter-side in same-type)
- interleaved vs speaker-specific side-by-side view

Deferred to a follow-up pass:

- starter-mechanism analysis (signed, starter-normalized AUC asymmetry across all conditions)
- Human-AI decomposition (`speaker_type` `speaker_is_starter` within Human-AI only)

Analysis completed: 2026-04-23 12:22:24.491549